

Foundation (Jalapeno)

Q1. Solve the inequality $|x| < 3$.

- (A) $(-\infty, -3) \cup (3, \infty)$
 (B) $(-3, 3)$
 (C) $(-\infty, -3] \cup [3, \infty)$
 (D) $[-3, 3]$
 (E) $(-3, 3)$

Q2. Solve the inequality $|2x - 1| \geq 5$.

- (A) $(-\infty, -2] \cup [3, \infty)$
 (B) $(-\infty, -2) \cup (3, \infty)$
 (C) $[-2, 3]$
 (D) $(-\infty, 2] \cup [3, \infty)$
 (E) $(-2, 3)$

Q3. Graph the solution set of $|x - 2| < 1$ on a number line.

- (A) $[1, 3]$
 (B) $(1, 3)$
 (C) $[1, 3)$
 (D) $(1, 3]$
 (E) $[-1, 1]$

Q4. The temperature outside is between -5 and 5 degrees Celsius. Express this as an absolute value inequality.

- (A) $|x| \geq 5$
 (B) $|x| \leq 5$
 (C) $|x - 5| \leq 0$
 (D) $|x + 5| \geq 0$
 (E) $|x| > 5$

Q5. Solve the inequality $|3x - 2| < 11$.

- (A) $(-3, 4)$
 (B) $(-3, 4]$
 (C) $[-3, 4)$
 (D) $[-3, 4]$
 (E) $(-\infty, -3) \cup (4, \infty)$

Q6. A company packages boxes with a weight between 10 and 15 kilograms. Express this as an absolute value inequality.

- (A) $|x - 12.5| \leq 2.5$
 (B) $|x| \leq 12.5$
 (C) $|x| \geq 12.5$
 (D) $|x - 10| \geq 5$
 (E) $|x - 15| \leq 5$

Q7. Solve the inequality $|x + 2| > 4$.

- (A) $(-\infty, -6) \cup (2, \infty)$
 (B) $(-6, 2)$
 (C) $(-6, 2)$
 (D) $(-\infty, -6]$
 (E) $[2, \infty)$

Q8. Graph the solution set of $|x| \geq 2$ on a number line.

- (A) $(-\infty, -2] \cup [2, \infty)$
 (B) $(-\infty, -2) \cup (2, \infty)$
 (C) $[-2, 2]$
 (D) $(-2, 2)$
 (E) $(-\infty, 2)$

Open Ended Questions (FRQ)

Q9. The distance between two cities is at least 150 miles. Express this as an absolute value inequality and graph the solution set on a number line. Explain your reasoning.**Q10.** A bakery sells a total of 250 loaves of bread per day. The number of whole wheat loaves, x , is within 50 of the number of white loaves. Express this as an absolute value inequality and solve for x . Explain the meaning of your solution in the context of the bakery.

Chef's Tips

- Emphasize the importance of understanding the definition of absolute value.
- Use number lines to visualize the solution sets of absolute value inequalities.
- Have students create and solve their own absolute value inequalities based on real-world scenarios.